Candidate's name/ Index number ……

THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT

FORM ONE TERMINAL EXAMINATION
032 CHEMISTRY
MARKING SCHEME MAY 2025

SECTION A (15 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
C	D	B	A	B	D	C	C	D	B

(01@ = 10 marks)

2.

List A	(i)	(ii)	(iii)	(iv)	(v)
List B	D	F	E	H	C

(01@ = 05 marks)

SECTION B (70 Marks)

3. (a) Matter is anything that has mass and occupies space 02½ marks

(b)

- All matter is made up of tiny discrete particles.
- The particles are in constant motion.
- The kinetic energy of a particle changes with temperature; the higher the temperature the faster the particle moves. 02½ @ = 07½ marks
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4. (a) Mixtures can be categorized as homogenous or heterogeneous.

A homogeneous mixture is one where the substances forming the mixture are totally mixed together uniformly. Examples include **solutions** of salts and sugars in water. 03 marks

A heterogeneous mixture is one where the substances forming the mixture remain separate and one substance is spread throughout the other as small particles, droplets, or bubbles. All **emulsions** and **suspensions** fall under this category. 03 marks

(b) An emulsion is a cloudy mixture of tiny droplets of one liquid suspended in another liquid. An emulsion forms when two immiscible liquids do not separate out into two layers when mixed together, and instead, one of the liquid forms droplets and spread throughout the other.

For example, cooking oil and water do not mix but they will form an emulsion when they are mixed and shaken, where droplets of oil spread throughout the water. Unlike pure liquids, emulsions are cloudy. 04 marks

5. (a) An element is a substance with contains the same type of atoms/An element is a single pure substance that cannot be broken down to anything simpler by a chemical change. 02 marks

(b) (i) The assigning of symbols for chemical elements followed the following rules:

- The symbol is the capitalized initial letter of the element. For example, the symbol for hydrogen is H and Potassium (Kalium) is K. 02 marks
- The symbol is the first capital letter plus the second lower case letter of the element. For example, the symbol for calcium is Ca and for sodium (Natrium) is Na. 02 marks
- The symbol is the first capital letter plus any other lower case letter of the element. For example, the symbol for Zinc is Zn and Copper (Cuprim) is Cu. 02 marks

(ii)

Name	Symbol
Lithium	Li 00½ mark
Sodium	Na 00½ mark
Potassium	K 00½ mark
Magnesium	Mg 00½ mark

6.

	Solution	Suspension
1	Homogenous	Heterogeneous
2	Clear/transparent	Opaque
3	Components separated by evaporation.	Components separated by filtration.
4	Particles completely dissolved	Particle separate on standing

(*Any four* 02½ @ = 10 marks)

7. (a) Give any four differences between a physical change and a chemical change

Physical change	Chemical change
(i) No new substance is formed	A new substance is formed
(ii) Can be reversed	The change is irreversible
(iii)No heat or only a little heat is released or absorbed	Heat is released or absorded
(iii) There is no change in mass	There is a change in mass

Any four 02@ = 08 marks

(b)

(i) The components of air can be separated by physical means …

(ii) The components of are not in a fixed ratio 02 marks

8. (a) the study of matter and their composition…. 02 marks

(b) (i) Chemistry is applied in other fields such as agriculture, manufacturing, medicine, processing and food industries, education, cosmetics and home care industries, etc. 04 marks

(ii)Materials made by the application of chemistry knowledge include soap, chalk, shoes, clothes, petroleum products, alcoholic and non-alcoholic beverages, cosmetics, drugs, and many others. 04 marks

9.

	Mixture (sea water)	Compound (Pure water)
1	The components of sea water (a mixture) can be separated by physical means, example, by distillation	The components of a compound can be separated by chemical means only.
2	The composition of a mixture can vary widely, example, sea water can contain various amount of salt and pure water.	Compounds are fixed in their compositions by mass of elements present, example, there are always 2 atoms of hydrogen to 1 atom of oxygen in a molecule of water
3	Formation of mixture is not usually accompanied by heat changes.	Formation of a compound is usually accompanied by heat changes.
4	Properties of a mixture are the sum of the properties of the individual constituents of the mixture.	The properties of a compound are entirely different from those of its constituent elements. For example, water is a liquid whereas its constituent elements, hydrogen and oxygen, are both gases.

$02½@ = 10$ marks

SECTION C (15 MARKS)

10. Introduction: the study of matter and their composition… (02 marks)

Chemistry is divided into various branches:

- Analytical Chemistry: Deals with studies and uses of instruments and methods used to separate, identify, and quantify chemical species in matter.
- Inorganic Chemistry: deals with properties and behavior of inorganic compounds.
- Organic Chemistry: Deals with carbon compounds
- Physical Chemistry: concerned with the relations between the physical properties of substances and their chemical compositions and transformations. (Any four $02½ @ = 10$ marks)
- Biochemistry: deals with chemical processes that occur in living things

In studying and applying the knowledge and skills of Chemistry, chemists have to acquire scientific skills in order to be successful in obtaining facts and verifying them. (02 marks)

Conclusion: Any relevant (01 mark)